

Date: 21 Jan 2025

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Digital Signal Processing

Lecture 1

Book: Digital Signal Processing

Chapter 2

Discrete Time Signal and Systems

Application of DSP

↳ Audio Processing

↳ Image Processing

↳ Speech Processing

↳ Communication

Signal series of data that is 0 dimensional

↳ Sensor and Control

↳ Video Processing

↳ Software define radio.

Topic 2.1

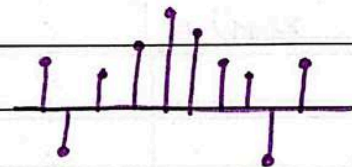
Discrete Time Signal

↳ Function of an independent variable that is an integer

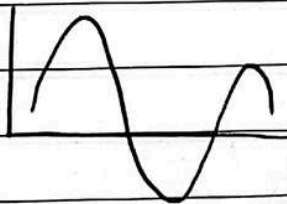
↳ does not define b/w two successive events

↳ Not analog (bcz it is not continuous)

define for every value of n ($-\infty < n < \infty$)

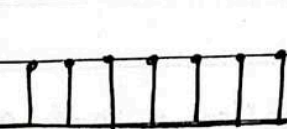


Analog
Signal
 $x_a(t)$



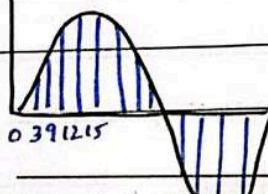
at every point of x there exist some value

Sampling
Pulse



value is taken at a particular event from analog signal

Sampled
Output



where T is sampling period (Time b/w successive samples/events)
⇒ Time period is 3 for this signal

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Sampling

- ↳ Under Sampling (bht kaam sample lg)
- ↳ Over sampling (bht zada sample lg)

Criteria for Sampling is Nyquist Criteria

Why do we want discrete signal?

We cannot process analog signal as it is ^{not} define its time period is infinite so complex and expensive so we discretise it we don't need to sample it more and less which will not map the actual analog signal

Ways of Representing Signal :

1, Graphical Representation:

2, Functional Representation:

$$x(n) = \begin{cases} 1 & \text{for } n = 1, 3 \\ 4 & \text{for } n = 2 \\ 0 & \text{elsewhere} \end{cases}$$

3, Tabular Representation:

n	-2	-1	0	1	2	3	4	5
x(n)	0	0	0	1	4	1	0	0

4, Sequence Representation: An infinite duration signal or sequence with the time origin (n=0)

Indicated by the symbol (↑) is represented as



$$x(n) = \{ \dots 0, 0, 1, 4, 1, 0, 0, \dots \}$$

↑

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Case where no negative number (arrow will be at most left)

condition $x(n) \geq 0$ for $n < 0$

$$x(n) = \{0, 1, 4, 1\}$$

↑

arrow
 $n=0$

$x(0)$ value where arrow is indicating

$$x(n) = \{3, -1, -2, 5, 0, 4\}$$

↑ ↑ ↑ ↑ ↑ ↑

$x(-2)$ $x(-1)$ $x(0)$ $x(1)$ $x(2)$ $x(3)$



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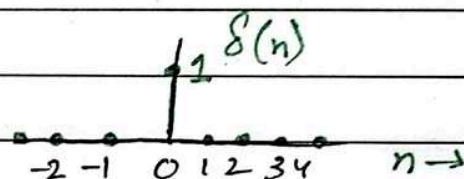
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Lecture 2

Topic 2.1.1 Some Elementary Discrete Time Signal

Unit Sample Signal $\delta(n)$

$$\delta(n) = \begin{cases} 1 & \text{for } n = 0 \\ 0 & \text{for } n \neq 0 \end{cases}$$

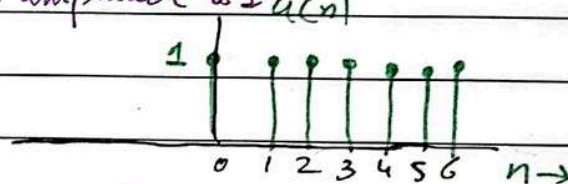


→ Zero everywhere except $n=0$

→ Signal is sometime Refer as Unit impulse signal

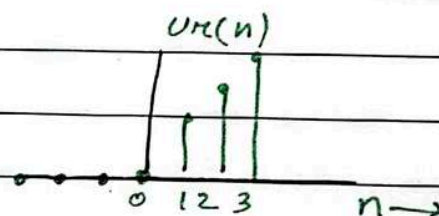
Unit Step signal $U(n)$ If any signal its amplitude is 1 $u(n)$

$$U(n) = \begin{cases} 1 & \text{for } n \geq 0 \\ 0 & \text{for } n < 0 \end{cases}$$



Unit Ramp Signal $U_r(n)$

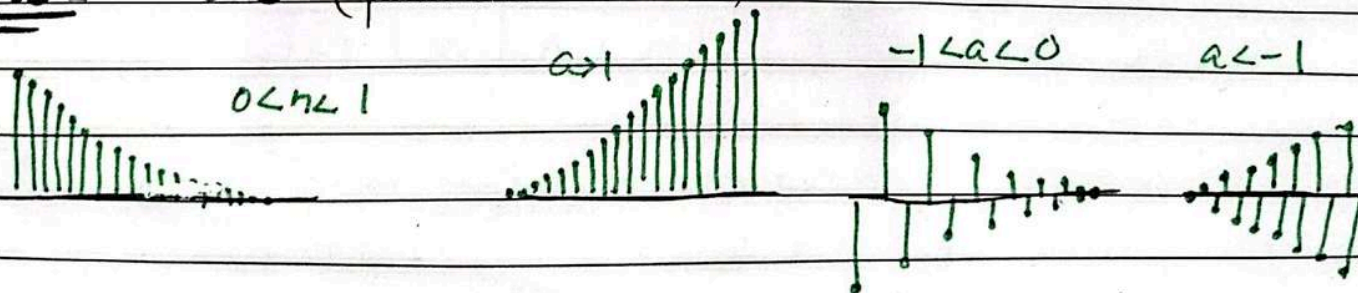
$$U_r(n) = \begin{cases} n & \text{for } n \geq 0 \\ 0 & \text{for } n < 0 \end{cases}$$



Exponential Signal

$$x(n) = a^n \quad \text{for all } n$$

Case 1 Real (parameter a is Real)



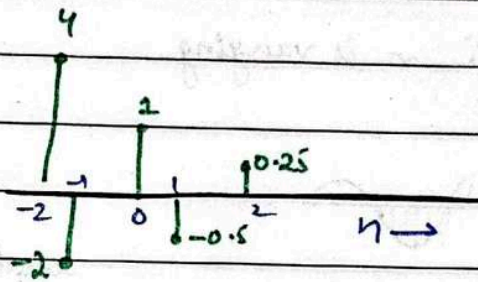
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$x(n) = a^n$ for all n
Let $n = -2:1:2$

$a = -0.5$

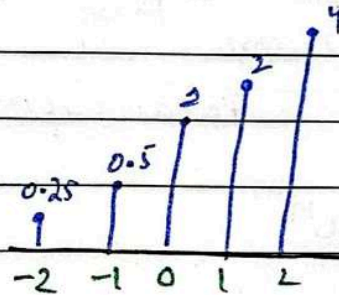
n	a^n
-2	4
-1	-2
0	1
1	-0.5
2	0.25



$y = a^n$ for all n
Let $a = 2$

$n = -2:1:2$

n	a^n
-2	0.25
-1	0.5
0	1
1	2
2	4



Case 2 Complex (a is complex value)

$$a = re^{j\theta}$$

$r \rightarrow$ amplitude $\theta \rightarrow$ angle

$$x(n) = a^n$$

$$x(n) = (re^{j\theta})^n = r^n e^{jn\theta}$$

$$x(n) = r^n e^{jn\theta}$$

$$x(n) = r^n (\cos \theta n + j \sin \theta n)$$

$$x(n) = r^n \cos \theta n + j r^n \sin \theta n$$



Real part of $x(n)$ Imaginary part of $x(n)$

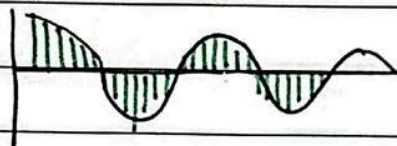
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$$x_R(n) = r^n \cos \theta n$$

$$x_I(n) = r^n \sin \theta n$$

$$r = 0.9 \quad \theta = \frac{\pi}{10} \quad n \text{ is varying}$$



Signal
 ↗
 Amplitude Angle

If we want to observe about the itself behaviour of amplitude and phase we will represent it in polar form

$$a^n$$

$$a^n = r^n e^{j\theta n} = \underbrace{|a|}_{\text{absolute value}} \angle \underbrace{\theta}_{\text{Phase}}$$

Polar Representation :

Amplitude Function:

$$|x(n)| = A(n) = r^n$$

Phase Function:

$$\angle x(n) = \phi(n) = \theta n \quad (\text{Linear relation with } n)$$

↳ vary with n

bec as n increase or decrease θ increase or decrease

Phase is either define from $-\pi$ to π or $-\pi$ to 2π bec these range are periodic After 2π it will repeat itself

So for our range and moreover if we don't define range it will be infinite

Process for Plotting θ b/w $-\pi$ to π

This process is called Modulo 2π

Modulo 2π (Normalise phase to $-\pi$ to π)

We subtract multiple of 2π from θn before plotting

this process is called modulo 2π This is just for normalise



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$$\phi_{\text{normalized}} = \phi(n) - 2\pi \text{round}\left(\frac{\phi(n)}{2\pi}\right)$$

without restricting this range the phase continue to grow indefinitely as n increases

Example

$$\theta(n) = \left\{ -3\pi, -\frac{\pi}{2}, \pi, \frac{3\pi}{2}, 3\pi \right\}$$

Range $(-3\pi \text{ to } 3\pi)$

when normalized range is $(-\pi \text{ to } \pi)$

$$= -3\pi - 2\pi \text{round}(-1.5)$$

$$= -3\pi - 2\pi(-1.5)$$

we round the

$$\text{Range of } \phi(n) = \left\{ -\pi, -\frac{\pi}{2}, \pi, \frac{3\pi}{2}, \pi \right\}$$

$$= -3\pi - 2\pi(-2)$$

$$= -3\pi + 4\pi$$

$$= \pi$$

Topic 2.1.2

Classification of Discrete Time Signal

Energy Signal & Power Signal

$$E = \sum_{n=-\infty}^{\infty} |x(n)|^2$$

Energy can be finite and infinite

Case 1 Finite Energy $\Rightarrow$
 $(0 < E < \infty)$

if E is finite that $x(n)$ is energy signal



Many Energy signal possess infinite energy and finite average power

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$$\text{Average power} = P = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x(n)|^2$$

Energy

If we define signal energy between finite interval $-N \leq n \leq N$

$$E_N = \sum_{n=-N}^N |x(n)|^2$$

Signal with finite interval

$$\text{Signal Energy} = E = \lim_{N \rightarrow \infty} E_N$$

Relation E & E_N

$$E = \lim_{N \rightarrow \infty} \sum_{n=-N}^N |x(n)|^2$$

$$\text{Average power of signal} = P = \lim_{N \rightarrow \infty} \frac{1}{2N+1} E_N$$

if E is finite power is zero

if E is infinite $\rightarrow$ power can be finite so it is power signal
 $\rightarrow$ power can be infinite

Example E is finite $P = 0$

$$E = \sum_{n=-N}^N |x(n)|^2$$



$$P = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x(n)|^2$$

energy

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$$P = \frac{1}{\infty} = 0$$

$$P = 0$$

Example 2.1.1 Determine power and energy of unit step sequence

↓
Amplitude 1 for $n \geq 0$

$$P = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=0}^N u^2(n)$$

$$\text{unit step } u(n) = \begin{cases} 1 & n \geq 0 \\ \text{elsewhere } 0 \end{cases}$$

$$\frac{1}{N} = \frac{1}{\infty} = 0$$

due to unit step signal

$$= \lim_{N \rightarrow \infty} \frac{N+1}{2N+1} = \lim_{N \rightarrow \infty} \frac{1 + \frac{1}{N}}{2 + \frac{1}{N}} = \frac{1}{2}$$

consequently it is a power signal

$$= \sum_{n=0}^N u^2(n)$$

Here $N=2$ (for example)

$$= u^2(0) + u^2(1) + u^2(2)$$

$$= (1)^2 + (1)^2 + (1)^2$$

$$= 3 \Rightarrow N+1$$

$$E = \sum_{n=-\infty}^{\infty} |u(n)|^2$$

$$E = \sum_{n=0}^{\infty} (1)^2$$

→ i.e. 1 so that unit step function does not exist. The extreme of this sum is infinity we will never find its answer so we will

$$E = \infty$$

consider energy infinite

Similarly if $x(n) = Ae^{j\omega n}$ its power is A^2
 Ramp is neither power nor energy signal



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Lecture 3

Periodic & Aperiodic Signal

Periodic:

Repeat itself after a fix time interval

Follow a same pattern

$N \rightarrow$ Time period (alsi time period ma ya dabara start hoga)

$N > 0$

$$x(n) = x(n+N) \text{ for all } n$$

Sine and cosine are periodic

The sinusoidal signal of the form

$x(n) = A \sin 2\pi f_0 n$ is periodic when f_0 is

rational number

$$f_0 = \frac{k}{N} \text{ where } k \text{ and } N \text{ both are integers}$$

Symmetric (even) and antisymmetric (odd) signal

→ Even signal same on both side

$$x(-n) = x(n)$$

→ Odd signal left and right opposite

$$x(-n) = -x(n)$$

If $x(n)$ is odd then $x(0) = 0$



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Topic 2.1.3

Simple Manipulations of Discrete Time signal

Time shifting takes place by replacing n to $n-k$
 if k is positive then signal is delayed by k unit
 if k is negative then signal is advanced by k units

Example 2.1.2

$$y(n] = x(n-3)$$

Step 1

$$n-3 \geq 0$$

$$n \geq 3$$

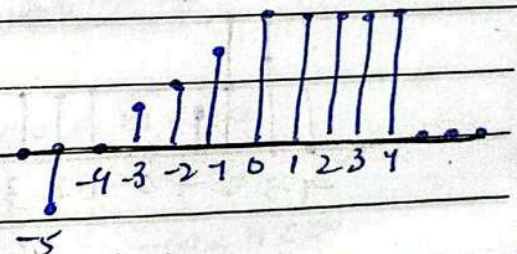
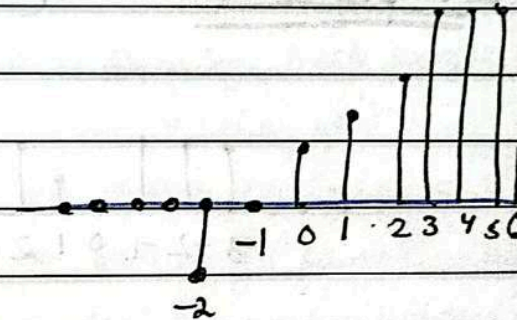
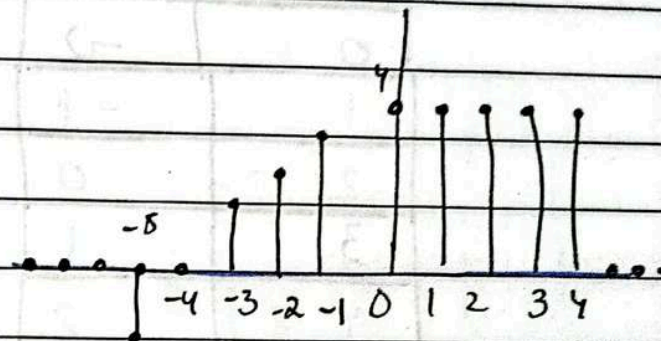
Step 2

Starting point $-5+3 = -2$ Ending $4+3 = 7$

Step 3

$x(n-3)$	$y(n)$
-5	-2
-4	-1
-3	0
-2	1
-1	2
0	3
1	4
2	5
3	6
4	7

These are indices not amplitude



Example

$$y(n] = x(n+2)$$

Step 1

$$n+2 \geq 0$$

$$n \geq -2$$

Step 2

Starting point $-5-2 = -7$

Ending point $4-2 = 2$

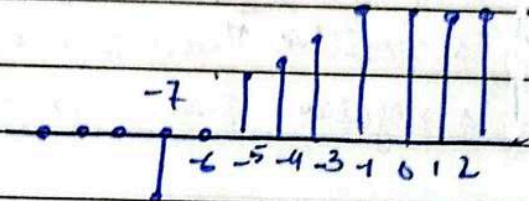


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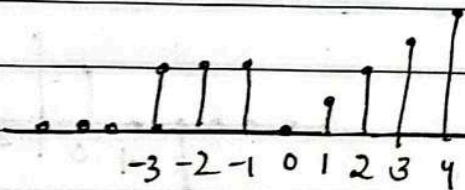
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$x(n+2)$	$y(n)$
-5	-7
-4	-6
-3	-5
-2	-4
-1	-3
0	-2
1	-1
2	0
3	1
4	2



Example 2.1.3

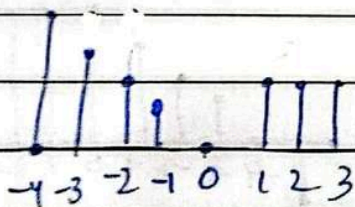
$x(-n)$ $x(-n+2)$



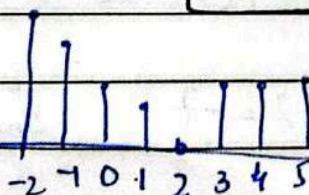
$$-n+2=0$$

$$n=2$$

$x(-n)$



$x(-n+2)$	$y(n)$
-4	-2
-3	-1
-2	0
-1	1
0	2
1	3
2	4
3	5



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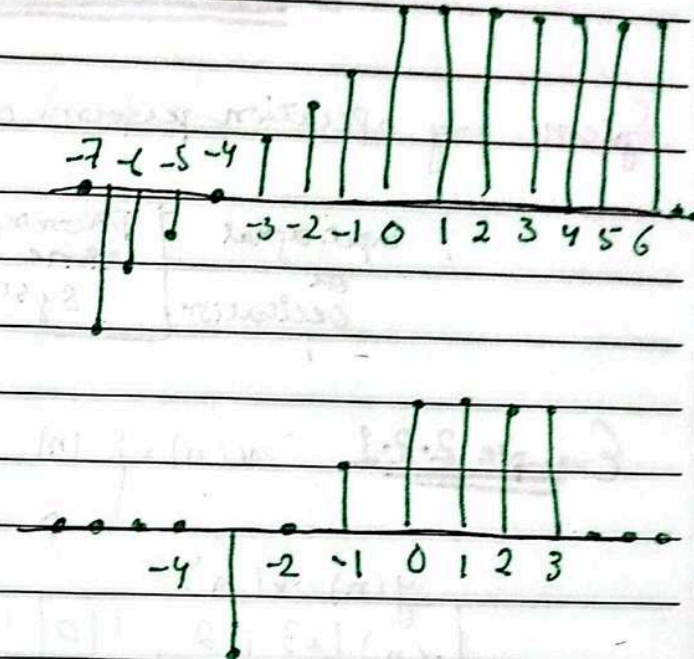
Example 2.1.4 $y(n) = x(2n)$

Step 1 $n \geq 0$

Step 2 $-7 \leq y(n) \leq 6$

Step 3

	$x(2n)$	$y(n)$
No value Hence discard	-14	-7
	-12	-6
	-10	-5
	-8	-4
	-6	-3
	-4	-2
	-2	-1
	0	0
	2	1
	4	2
	6	3
No value Hence discard	8	4
	10	5
	12	6



→ Down sampling (Kuch sample remove ho jaway ga)
 Kuch values nahi aaway gi

→ Over sampling (sample remain same but more values is added)

Addition, multiplication and scaling of sequence

→ Amplitude scaling

$$y(n) = A x(n) \quad -\infty < n < \infty$$

→ Sum

$$y(n) = x_1(n) + x_2(n) \quad -\infty < n < \infty$$

→ product

$$y(n) = x_1(n) x_2(n) \quad -\infty < n < \infty$$



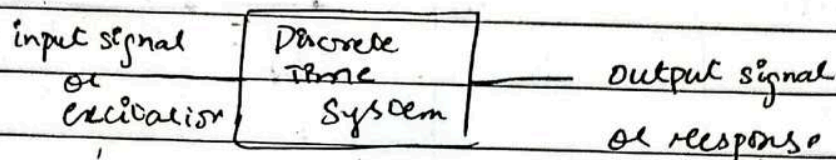
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Topic 2.2

Discrete Time Systems

System any operation perform on signal that generate $y(n)$



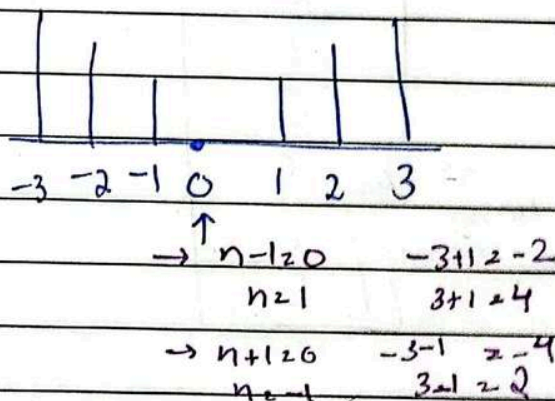
Example 2.2.1

$$x(n) = \begin{cases} |n| & -3 \leq n \leq 3 \\ 0 & \text{otherwise} \end{cases}$$

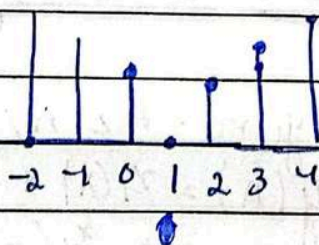
$$y(n) = x(n)$$

$x(n)$	+3	2	1	0	1	2	3
n	-3	-2	-1	0	1	2	3

$x(n-1)$
 ↳ shift leftward
 $x(n+1)$
 ↳ shift rightward



$x(n-1)$	$y(n)$
-3	-2
-2	-1
-1	0
0	1
1	2
2	3
3	4



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$$d) y(n) = \frac{1}{3} [x(n+1) + x(n) + x(n-1)]$$

at $n=0$

$$y(0) = \frac{1}{3} [x(0+1) + x(0) + x(0-1)]$$

$$= \frac{1}{3} [x(1) + x(0) + x(-1)]$$

$$= \frac{1}{3} [1 + 0 + 1] = \frac{2}{3}$$

$$e) y(n) = \max [x(n+1), x(n), x(n-1)]$$

at $n=0$

$$y(0) = \max [x(0+1), x(0), x(0-1)]$$

$$= \max [x(1), x(0), x(-1)]$$

$$= 1$$

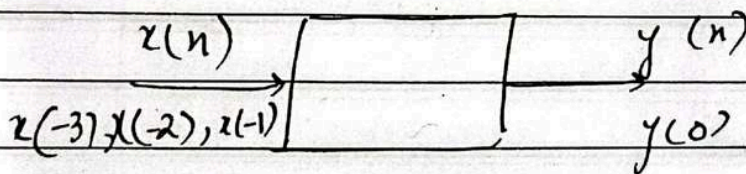
max value is 1 so $x(0)$ is at 1

$$f) y(n) = \sum_{k=-\infty}^n x(k) = x(n) + x(n-1) + x(n-2) + \dots$$

↳ Accumulator (previous all values ka sum)

↳ (addition of past values take place)
 $-\infty$ to n all values ko add

$(n+1)$ is prediction



$$y(n) = \sum_{k=-\infty}^n x(k) = \sum_{k=-\infty}^{n-1} x(k) + x(n)$$

present value
is separate

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Example 2.2.2 $x(n) = n u(n)$

a) $y(-1) = 0$

$$y(n) = \sum_{k=-\infty}^n x(k)$$

$$= \sum_{k=-\infty}^{-1} x(k) + \sum_{k=0}^n x(k)$$

$$= 0 + \frac{n(n+1)}{2}$$

$$\sum_{k=0}^n x(k) = \frac{n(n+1)}{2} \quad n \geq 0$$

b) $y(-1) = 1$

$$y(n) = \sum_{k=-\infty}^n x(k)$$

$$= \sum_{k=-\infty}^{-1} x(k) + \sum_{k=0}^n x(k)$$

$$= 1 + \frac{n(n+1)}{2}$$

$$= 1 + \frac{n^2 + n}{2}$$

$$\boxed{y(n) = \frac{n^2 + n + 2}{2} \quad n \geq 0}$$

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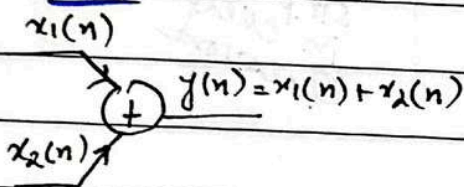
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Lecture 4

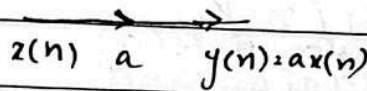
Topic 2.2.2

Block diagram representation of discrete time system

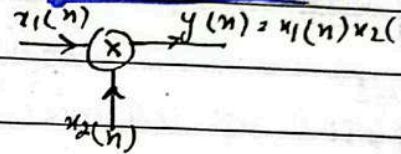
Adder



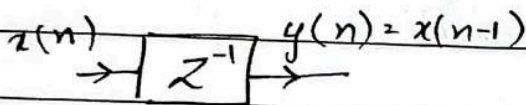
Constant Multiplier



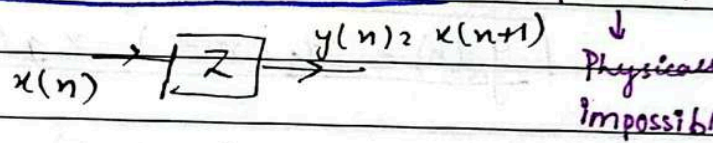
Signal Multiplier



Unit delay element

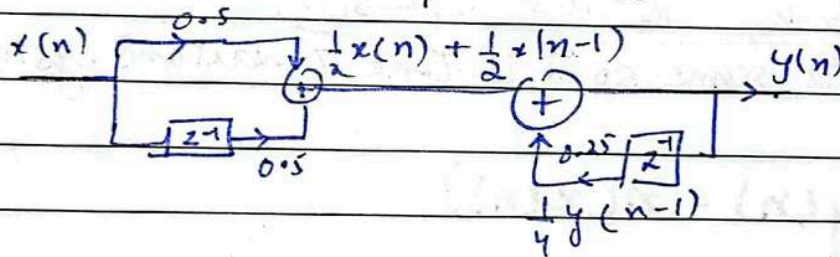


Unit advancement element (predictor)



Example 2.2.3

$$y(n) = \frac{1}{4} y(n-1) + \frac{1}{2} \left(x(n) + \frac{1}{2} x(n-1) \right)$$



Signal
 ↓
 Information

System
 ↓
 operation perform on signal to generate output

Topic 2.2.3 Classification of Discrete Time system

Static versus dynamic system

→ don't need more data (no storage need)

Static system → memoryless (not past & future sample)

→ depend on present value (output & input is same)

Dynamic system → memory

→ give good output but more data → depend on future or past
 saved storage to generate output

Time variant versus Time invariant system

Time variant Input and output vary with time

$N=0 \rightarrow$ static

$0 < N < 1$ if $N \neq 1$

Osa peechaynaki ja sakata but on past

Time invariant Input & output does not change with time

$$\begin{matrix} x(n) & \xrightarrow{P} & y(n) \\ \vdots & & \vdots \\ x(n-k) & & y(n-k) \end{matrix}$$

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Example 2.24 (a) $y(n) = x(n) - x(n-1]$ (dynamic)

Step 1 $\rightarrow$ b¹ equation ko aisko change kar ky x_1 & y_1 ma

$$y_1(n) = x_1(n) - x_1(n-1) \rightarrow (1)$$

Step 2 introduce delay in $x_1 \rightarrow$ original

$$x_2(n) = x_1(n-k)$$

Step 1 of LTI (we introduce time delay in input)

As x_2 is new input so it give new output

$$y_2(n) = x_2(n) - x_2(n-1)$$

$$y_2(n) = x_1(n-k) - x_1(n-1-k) \rightarrow (2)$$

Step 2 of LTI (introduce delay in output)

$$y_1(n) = x_1(n) - x_1(n-1)$$

$$\Rightarrow n = n-k$$

$$y_1(n-k) = x_1(n-k) - x_1(n-k-1) \rightarrow (3)$$

As eq (2) & (3) is same so it is time invariant system

Example 2.24 (b) $y(n) = n(x(n))$

Step 1
delay in
input
of LTI

$$y_1(n) = n x_1(n) \rightarrow (1)$$

introduce time delay in input

$$x_2(n) = x_1(n-k) \rightarrow$$

$$y_2(n) = n x_2(n)$$

$$y_2(n) = n x_1(n-k) \rightarrow (2)$$

Step 2
delay
in
output

$$y_1(n) = n x_1(n) \Rightarrow n = n-k$$

$$y_1(n-k) = (n-k) x_1(n-k) \rightarrow (3)$$

eq (2) & (3) is not same so time variant system



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Example 2.2.4(c)

$$y(n) = x(-n)$$

Step 1
delay in input

$$\left[\begin{array}{l} y_1(n) = x_1(-n) \rightarrow (1) \\ x_2(n) = x_1(n-k) \\ y_2(n) = x_2(-n) \\ \underline{y_2(n) = x_1(-n-k)} \end{array} \right] \rightarrow (2)$$

Step 2
delay in output

$$\left[\begin{array}{l} y_1(n) = x_1(-n) \quad n = n-k \\ y_1(n-k) = x_1(-(n-k)) \\ \underline{y_1(n-k) = x_1(-n+k)} \end{array} \right] \rightarrow (3)$$

it is time variant

Linear system
is which holds
superposition

Linear versus non linear system

Superposition
/ \
additivity multiplicative

Linear system follow superposition principle

$$\hookrightarrow T[a_1 x_1(n) + a_2 x_2(n)] = a_1 T[x_1(n)] + a_2 T[x_2(n)]$$

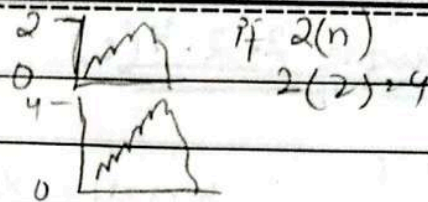
$$\hookrightarrow T[a_1 x_1(n)] = a_1 T[x_1(n)]$$



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Lecture 5



Example 2.26

a)

b)



Include all previous values

Causal System



Present and Past value

c) $y(n) = n(x(n))$ here we obtain the

Non Causal System

Original signal only its amplitude is increase $\downarrow$

g) If n is ∞ then we will obtain future value so (non causal)

Future Value

Any future value present in system will result in non causal system

Stable & Unstable System

Stable System

Bounded input it produces a bounded output

Bounded values should not go to infinity

$$\text{Input} \leftarrow \{x(n)\} \leq M_x < \infty$$

$$\text{Output} \leftarrow \{y(n)\} \leq M_y < \infty$$

Input and output goes to some value but less than ∞

Any value

Example 2.27

$$y(n] = y^2(n-1) + x(n)$$

where

$$x(n) = C \delta(n)$$

assume

$$y^2(-1) = 0$$

If $n=0$

$$y(0) = y^2(-1) + x(0)$$

$$y(0) = C \rightarrow 0 + C(1) = C$$

$$0 + C = C$$

impulse only exist at $n=0$



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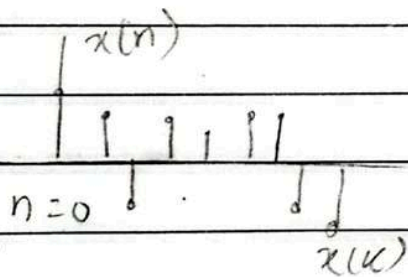
$$\begin{aligned}
 y(1) &= y^2(0) + x(1) \\
 &= (c)^2 + 0 \\
 \boxed{y(1) = c^2}
 \end{aligned}$$

Topic 2.3.2

Resolution of Discrete Time signals in sum of impulse signals

$$n = k$$

$$x(n) \delta(n-k) = x(k) \delta(n+k)$$



- If $n=0$ then it will remain at same position
- If $n-k$ then shifted right ward
- If $n+k$ then shifted left ward

$\delta(n)$	$\delta(n-k)$	
$n = 0, 1, 2$	here $n=2$ $k=3$	$\left\{ \begin{array}{l} \text{if } n=2=k \\ \delta(0) \end{array} \right.$
	$\delta(2-3) = \delta(-1)$	

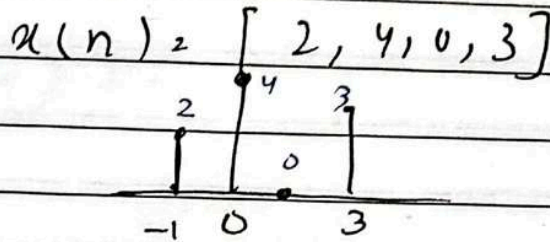


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Example 2.3.1

Impulse unit
area is 1



here no impulse exist
if we want to write this
function in the form of
impulses then we will do

$$2\delta(n+1) + 4\delta(n) + 0\delta(n-1) + 3\delta(n-3)$$

This impulse
equal to 1

impulse multiplies

Convolution sum

$$y(n) = \sum_{k=-\infty}^{\infty} x(k)h(n-k)$$

$$w(k) = x(k)h(-k)$$

if we want to convolve a signal
firstly we need to find $h(-k)$

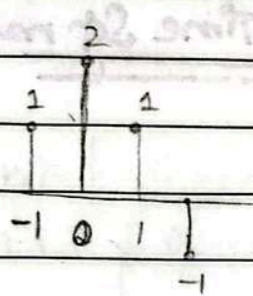
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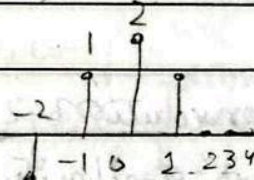
Example 2.3.2

$$h(n) = [1, 2, 1, -1]$$

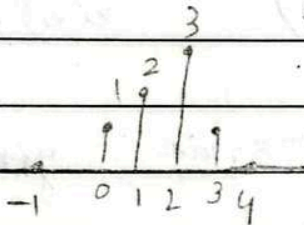
$h(n) =$
 $h(k)$



Step 1 convert $h(n)$ into $h(-n)$
 $h(-k)$



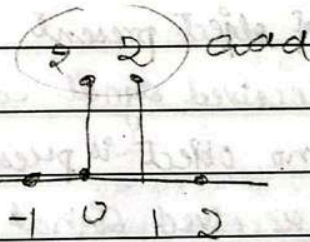
$x(n)$



Step 2 multiply each value of $x(k)$ with $h(-k)$

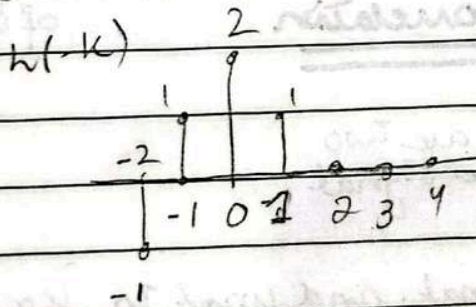
This will provide us value of $y(0)$

This will provide 0th sample at output

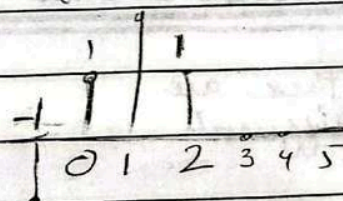


Now add these for value
for $y(0) = 2 + 2 = 4$
 $y(0) = 4$

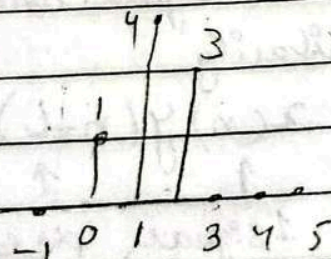
$h(1-k)$



$h(1-k)$



$$y(1) = 4 + 3 + 1 = 8$$



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Lecture 6

Topic 2.3 Correlation of Discrete Time Signal

→ Similar to convolution

→ Use to observe similarity b/w two signals

$$y(n) = \underbrace{a}_{\text{Scaled}} \underbrace{(x-D)}_{\text{Delay}} + \underbrace{W(n)}_{\text{additive noise}}$$

$x(n) \rightarrow$ Transmitted signal

$y(n) \rightarrow$ Received signal

In case if object present

Received signal contain noise, delay, scaled

In case no object is present

Received signal is noise there is too much noise in the signal we

are not getting whether it is delay signal, noisy or scaled signal

$$y(n) = \underbrace{x_i(n)}_{x_0(n) \cdot x_1(n)} + W(n) \quad \begin{matrix} i=0,1 \\ 0 \leq n \leq L-1 \end{matrix}$$

Topic 2.6.1

Crosscorrelation and autocorrelation

↓
Means there are two different signal

↓
These are two same signal

↓
digit display number of samples

Cross correlation we take 2 different signal and want to know about their similarity

$$r_{xy}(l) = \sum_{n=-\infty}^{\infty} x(n) y(n-l) \quad l=0, \pm 1, \pm 2, \dots$$

↑
shows which sequences of signal are multiplied

↑
1 signal delay version of 2 signal



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if we take $n = -5$ to 5 there are total 10 sample
 firstly we will take $n = 0$ where we observe behaviour of 2 non
 delayed signal

Autocorrelation

Example

$x(n) = \{ \dots, 0, 0, 2, -1, 3, 7, 1, 2, -3, 0, 0, \dots \}$
 $y(n) = \{ \dots, 0, 0, 1, -1, 2, -2, 4, 1, -2, 5, 0, 0, \dots \}$
 we take 2 similar
 signal and observe
 other signal is same
 as input signal

$$R_{xy}(l) = \sum_{n=-\infty}^{\infty} x(n) y(n-l)$$

$$R_{xx}(l) = \sum_{n=-\infty}^{\infty} x(n) x(n-l)$$

$$= \sum_{n=-4}^3 x(n) y(n-0)$$

$$= \sum_{n=-4}^3 x(n) y(n)$$

$$= x(-4)y(-4) + x(-3)y(-3) + x(-2)y(-2) +$$

$$x(-1)y(-1) + x(0)y(0) + x(1)y(1) + x(2)y(2) + x(3)y(3)$$

$$= (2)(1) + (-1)(-1) + 3(2) + 7(-2) + 1(4) + 2(1) + (-3)(-2) + 0$$

$$= 2 + 1 + 6 - 14 + 4 + 2 + 6$$

$$= 7$$

we have to select value of
 n we need to be carefully
 than we don't miss any
 value of both signal
 we need to see the range
 of both signals

In full signal after
 cross correlation
 the highest value is
 the point where
 both signals are
 maximum correlation



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rational $\rightarrow$ w/o
number

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$\rightarrow$ in real life signal it is difficult to deal with
the domain signal we cannot get enough information

Chapter 4 Frequency

We decompose the signal in terms of
sinusoidal (or complex exponential) signal

Fourier transform
 $\downarrow$
useful in analysis
and design of
LTI system

Topic 4.2 Frequency analysis of discrete time signal

Any wave can be generated by addition
of harmonics

Topic 4.2.1

Any periodic signal
can be represented

behaviour of exponent

$$\sum_{n=0}^{N-1} e^{j2\pi kn/N} = \begin{cases} N & k=0, \pm N, \pm 2N, \dots \\ 0 & \text{otherwise} \end{cases}$$

$k=0, \pm N, \pm 2N, \dots$
otherwise

$$x(n) = \sum_{k=0}^{N-1} c_k e^{j2\pi kn/N}$$

$\rightarrow$ k should be multiple of N or equal to N
 $\rightarrow$ if k is not equal to N or multiple of
 N then summation is 0

$\rightarrow$ $\sum$ is equal to N This contains
cos and sin we need to
select specific sequence
through which we can get
any signal
here we selected such wave
as cos. so it is limited
and sin remain and
sequence of sin gives
us rectangular signal

$$\sum_{n=0}^{N-1} e^{j2\pi kn/N} \cdot e^{-j2\pi ln/N}$$

$$= e^{j2\pi (k-l)n/N} = e^{j2\pi (k-l)n}$$

behaviour of exponent

$$\sum_{n=0}^{N-1} e^{j2\pi (k-l)n/N} = \begin{cases} N & \text{when } k=l \\ 0 & k \neq l \end{cases}$$

when $k=l$
 $k \neq l$



$k=l$

$$e^{j2\pi (0)n/N} = N$$

$k \neq l$

$$e^{j2\pi (k-l)n/N} \Rightarrow \alpha^n$$

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$$\sum_{n=0}^{N-1} x(n) e^{j 2 \pi \frac{(k-l)n}{N}} = \sum_{n=0}^{N-1} \sum_{l=0}^{N-1} C_k e^{j 2 \pi \frac{(k-l)n}{N}}$$

$C_k \quad N$
 $k=l$

$$\sum_{n=0}^{N-1} x(n) e^{j 2 \pi \frac{k n}{N}} = N C_k$$

$$\left| \frac{1}{N} \sum_{n=0}^{N-1} x(n) e^{j 2 \pi \frac{k n}{N}} = C_k \right|$$

whatever input you have
 you can find this C_k

$$\cos \frac{\pi}{3}$$

$$W = \frac{\pi}{3}$$

$$W = 2 \pi f$$

$$\frac{\pi}{3} = 2 \pi f$$

$$\frac{1}{6} = f$$

$$N = 6$$

$$\cos \theta = \frac{e^{j\theta} + e^{-j\theta}}{2}$$



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Lecture 7

Convolution

Example: Determine the output $y(n)$ of a relaxed LTI system with impulse response

$$h(n) = a^n u(n) \quad (|a| < 1)$$

cause decreasing behaviour $-1 < a < 1$

→ we want this

when the input is a unit step sequence that is

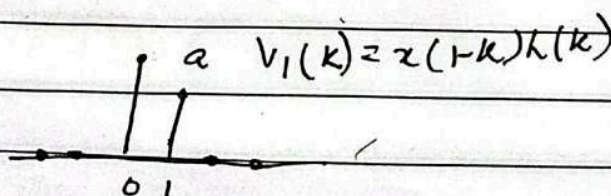
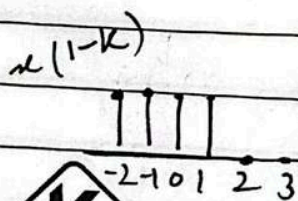
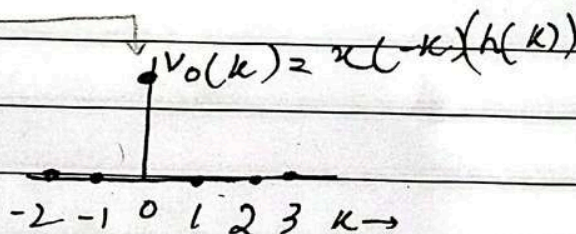
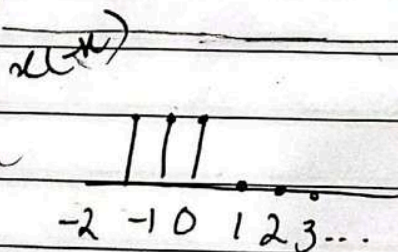
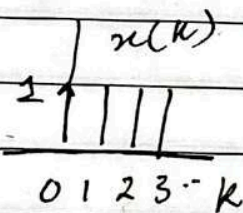
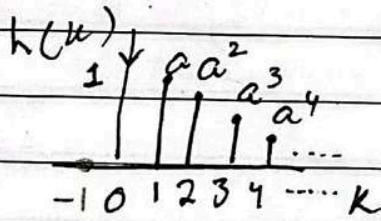
$$x(n) = u(n)$$

→ unit sequence (exist from zero to onwards)

Solution: In this case both $h(n)$ and $x(n)$ are infinite duration sequence we use following form of convolution formula

$$y(n) = \sum_{k=-\infty}^{\infty} x(n-k) h(k)$$

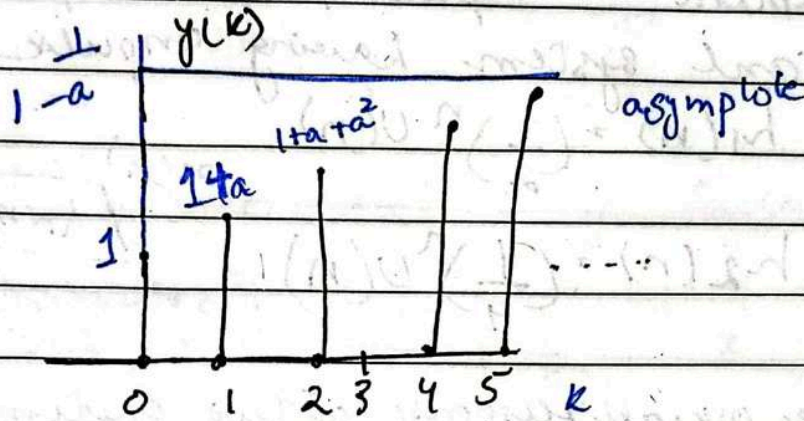
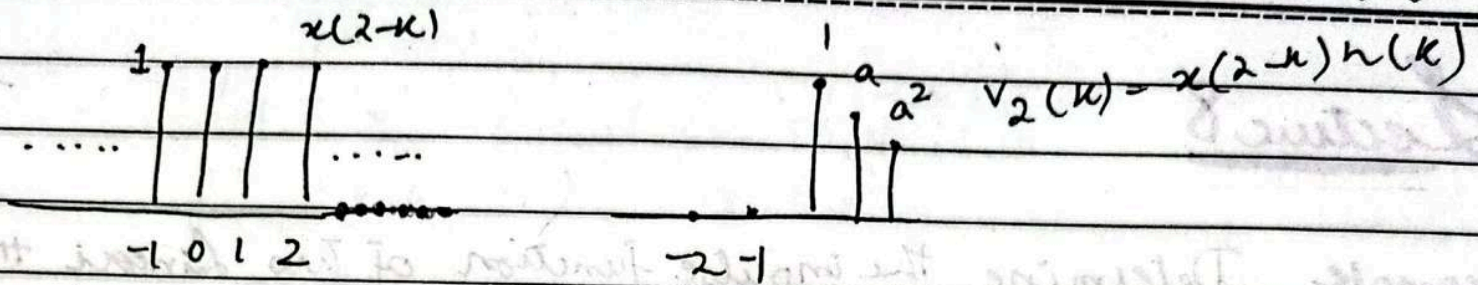
in which $x(k)$ is folded



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Case straight line ko curve approach kar hai ha ya tab touch karay gi jab n ya k ∞ to approach karay

by
 $y(0) = 1$
 $y(1) = 1+a$
 $y(2) = 1+a+a^2$
 $\vdots$

V_0 par jo amplitude
 V_1 par jo amplitude add

$$y(n) = 1 + a + a^2 + \dots + a^n$$

$$= \frac{1 - a^{n+1}}{1 - a} \Rightarrow \text{Geometric sum}$$

$$y(\infty) = \lim_{n \rightarrow \infty} y(n) = \frac{1}{1-a}$$

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Lecture 8

Example Determine the impulse function of two Linear time invariant system having impulse response

$$h_1(n) = \left(\frac{1}{2}\right)^n U(n)$$

$$h_2(n) = \left(\frac{1}{4}\right)^n U(n)$$

x(n) → [h₁(n)] [h₂(n)] → y(n)
 → shifted and fold

To determine the overall response of two systems in cascade we simply convolve $h_1(n)$ and $h_2(n)$ hence

$$h(n) = \sum_{k=-\infty}^{\infty} h_1(k) h_2(n-k)$$

where $h_2(n)$ is folded and shifted.

We define the product sequence

$$\begin{aligned} v_n(k) &= h_1(k) h_2(n-k) \\ &= \left(\frac{1}{2}\right)^k \left(\frac{1}{4}\right)^{n-k} \end{aligned}$$

$v(n) \& h(n)$
 is similar
 only difference
 $h(n)$ is summation

which is non zero for $k \geq 0$ and $n-k \geq 0$

On the other hand, for $n < 0$ we have $v_n(k) = 0$ for all k

Hence,

$$h(n) = 0 \quad n < 0$$

↓
 bit unit step function

For $n \geq k \geq 0$

$$h(n) = \sum_{k=0}^n \left(\frac{1}{2}\right)^k \left(\frac{1}{4}\right)^{n-k}$$

due to $U(n)$ for unit step. Pers has zero is zero

$$\therefore \left(\frac{1}{4}\right)^{n-k} = \left(\frac{1}{2}\right)^{2(n-k)} = \left(\frac{1}{2}\right)^{2(n-k)}$$



$$= \sum_{k=0}^n \left(\frac{1}{2}\right)^k \left(\frac{1}{2}\right)^{2(n-k)}$$

$$= \sum_{k=0}^n \left(\frac{1}{2}\right)^k \left(\frac{1}{2}\right)^{2n-2k}$$

Since $\left(\frac{1}{2}\right)^{2n}$ is independent of $\sum_k$ factor it out

$$= \left(\frac{1}{2}\right)^{2n} \sum_{k=0}^n \left(\frac{1}{2}\right)^k \left(\frac{1}{2}\right)^{-2k}$$

$$= \left(\frac{1}{2}\right)^{2n} \sum_{k=0}^n \left(\frac{1}{2}\right)^{-k}$$

$$\therefore \sum_{k=0}^n \left(\frac{1}{2}\right)^{-k} = \frac{1-2^{n+1}}{1-2} = 2^{n+1}-1$$

$$= \left(\frac{1}{2}\right)^{2n} (2^{n+1}-1)$$

$$= \left(\frac{1}{2}\right)^n \left(\frac{1}{2}\right)^n (2^{n+1}-1)$$

$$= \left(\frac{1}{2}\right)^n \left(\frac{1}{2}\right)^n (2^n \cdot 2 - 1)$$

$$= \left(\frac{1}{2}\right)^n [2^{-n} \cdot 2^n - 2^{-n}]$$

$$h(n) = \left(\frac{1}{2}\right)^n \left[2 - \left(\frac{1}{2}\right)^n \right]$$

Lecture 9

Power Density spectrum of Periodic Signal (PSD)

the spectrum that shows how dense the signal is

periodic

↳ repeat itself

after specific interval

If answer is not given def most sequence $n \geq 0$

$$P_x = \frac{1}{N} \sum_{n=0}^{N-1} |x(n)|^2$$

conjugate

$$x(n) = [1 -2 \ 1 -2 \ 1 -2]$$

$$P_x = \frac{1}{N} \sum_{n=0}^{N-1} x(n) x^*(n)$$

$$\frac{1}{2} \sum_{n=0}^1 |x(n)|^2$$

$$= \frac{1}{N} \sum_{n=0}^{N-1} x(n) \sum_{k=0}^N c_k^* e^{-j2\pi kn/N}$$

$$\frac{1}{2} [|x(0)|^2 + |x(1)|^2]$$

$$= \sum_{k=0}^N c_k^* \left[\frac{1}{N} \sum_{n=0}^{N-1} x(n) e^{-j2\pi kn/N} \right]$$

$$\frac{1}{2} [1^2 + |-2|^2]$$

$$= \sum_{k=0}^N c_k^* c_k$$

$$\frac{1}{2} [1 + 4] = \frac{5}{2}$$

$$P_x = \sum_{k=0}^N |c_k|^2$$

The average power is the sum of average power frequency component

Parseval's Theorem

$$P_x = \sum_{k=0}^N |c_k|^2$$

Energy of sequence $x(n)$

$$E_N = \sum_{n=0}^{N-1} |x(n)|^2 = N \sum_{n=0}^{N-1} |c_k|^2$$



Example 4.2.2

$$C_k = \frac{1}{N} \sum_{n=0}^{N-1} x(n) e^{-j \frac{2\pi}{N} kn} = \frac{1}{N} \sum_{n=0}^{L-1} A e^{-j \frac{2\pi}{N} kn}$$

repeat itself after L intervals
k=0, 1, 2, ...

amplitude of x(n)

$$C_k = \frac{A}{N} \sum_{n=0}^{L-1} \left(e^{-j \frac{2\pi}{N} k} \right)^n = \begin{cases} \frac{AL}{N} & k=0 \\ \frac{A}{N} \frac{1 - e^{-j \frac{2\pi}{N} kL}}{1 - e^{-j \frac{2\pi}{N} k}} & k=1, 2, \dots \end{cases}$$

scale for values other than zero

$$\sum_{n=0}^{N-1} \alpha^n = \sum_{n=0}^{\infty} \alpha^n - \sum_{n=N}^{\infty} \alpha^n = \frac{1}{1-\alpha} - \frac{\alpha^N}{1-\alpha}$$

$$\frac{1 - e^{-j \frac{2\pi}{N} kL}}{1 - e^{-j \frac{2\pi}{N} k}} = \frac{e^{-j \frac{2\pi}{N} kL}}{e^{-j \frac{2\pi}{N} k}} \left[\frac{e^{j \frac{2\pi}{N} kL} - e^{-j \frac{2\pi}{N} kL}}{e^{j \frac{2\pi}{N} k} - e^{-j \frac{2\pi}{N} k}} \right]$$

$$= e^{-j \frac{2\pi}{N} k(L-1)} \left[\frac{e^{j \frac{2\pi}{N} kL} - e^{-j \frac{2\pi}{N} kL}}{e^{j \frac{2\pi}{N} k} - e^{-j \frac{2\pi}{N} k}} \right]$$

$$= e^{-j \frac{2\pi}{N} k(L-1)} \left[\frac{\sin(\pi kL/N)}{\sin(\pi k/N)} \right]$$

$$C_k = \begin{cases} \frac{AL}{N} \\ \frac{A}{N} e^{-j \frac{2\pi}{N} k(L-1)} \left[\frac{\sin(\pi kL/N)}{\sin(\pi k/N)} \right] \end{cases}$$

Now all is written in form of sine instead of exponential

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Topic 4.2.3

The Fourier transform of discrete time aperiodic signals

for 1 period only

Fourier transform $X(\omega) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n}$

To check periodic with 2π

$$\omega = \omega + 2\pi$$

$$X(\omega + 2\pi k) = \sum_{n=-\infty}^{\infty} x(n) e^{-j(\omega + 2\pi k)n}$$

$$= \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n} e^{-j2\pi kn}$$

1 b/c $\cos 2\pi = 1$
 since

Synthesis equation
 Inverse transform $x(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\omega) e^{j\omega n} d\omega$

Fourier transform $X(\omega) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n}$

Topic 4.2.5

$$E_{xx} = \sum_{n=-\infty}^{\infty} |x(n)|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |X(\omega)|^2 d\omega$$

→ Parseval theorem of discrete time aperiodic signal



$$S_{xx} = |X(\omega)|^2$$

→ energy as function of frequency

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Example 4.2.3

$$x(n) = a^n u(n) \quad -1 < a < 1$$

$$\sum_{n=0}^{\infty} a^n = \frac{1}{1-a}$$

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$$x(n) = \begin{cases} a^n u(n) & |a| < 1 \\ -1 > a < 1 \end{cases}$$

$$X(\omega) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n}$$

$$= \sum_{n=0}^{\infty} a^n e^{-j\omega n}$$

$$= \sum_{n=0}^{\infty} (ae^{-j\omega})^n$$

$$\therefore \sum_{k=a}^{\infty} x^k = \frac{x^a}{1-x}$$

$$| | =$$

$$\sqrt{(\text{real})^2 + (\text{imag})^2}$$

for finding minima &
 maxima we take
 derivative

$$\angle X(\omega) = \angle_{\text{num}} - \angle_{\text{den}}$$



KARIMI
 NOTE

Date: _____

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Example 7.2.4

$$x(n) = \begin{cases} A & 0 \leq n < N-1 \\ 0 & \text{elsewhere} \end{cases}$$

$$X(\omega) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n}$$

$$= \sum_{n=0}^{N-1} A e^{-j\omega n}$$